underlying file system store. Figure 3.8 shows the inner workings of a Lucene engine during indexing.

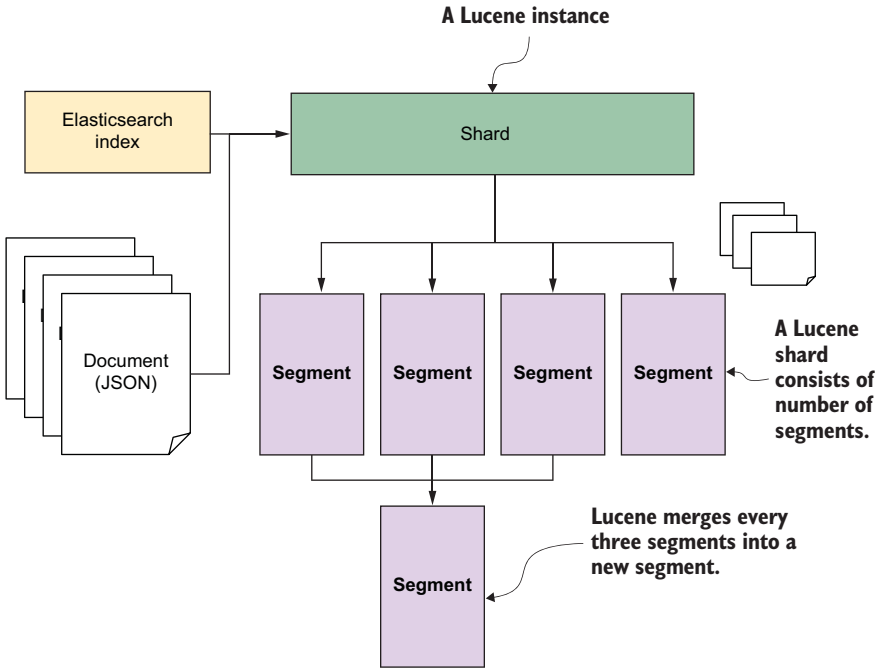


Figure 3.8 Lucene's mechanism for indexing documents

Shards are distributed across the cluster for availability and failover. On the other hand, replicas allow redundancy in the system. Once an index is in operation, shards cannot be relocated, as doing so would invalidate the existing data in an index.

As duplicate copies of shards, replica shards provide redundancy and high availability in an application. By serving read requests, replicas enable distributing the read load during peak times. A replica is not co-located on the same node as the respective shard, as that would defeat the purpose of redundancy. For example, if the node crashed, you would lose all the data from the shard and its replica if they were co-located. Hence shards and their respective replicas are distributed across different nodes in the cluster.

DEFINITIONS A *cluster* is a collection of nodes. A *node* is an instance of an Elasticsearch server. For example, when we start an Elasticsearch server on our machine, we are essentially creating a node. This node joins a cluster by default. Because the cluster has only this node, it is called a *single-node cluster*. If we start more instances of the server, they will join this cluster, provided the settings are correct.